MIL-A-47280(MI)
9 August 1974
SUPERSEDING
MIS-14110A
23 August 1965
MIS-13262
5 September 1967
MIS-12836
7 February 1967

MILITARY SPECIFICATION

ADHESIVES, EPOXY

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of room temperature curing, epoxy adhesives.

1.2 Classification. The adhesive shall be of the following types as specified:

Type I - High viscosity

Type II - Low viscosity

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-C-96	Cans, Metal, 28 Gage and Lighter

STANDARDS

Federal

FED-STD-406	Plastics: Methods of Testing

FSC 8040

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Military

MIL-STD-129 Marking for Shipment and Storage

(copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents for a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

American Society for Testing Materials

ASTM D257	Method of Test for Electrical Resistance of Insulating Materials
ASTM D638	Method of Test for Tensile Properties of Plastics
ASTM D695	Method of Test for Compressive Properties of Rigid Plastics
ASTM D901	Method of Testing Askarels
ASTM D1002	Method of Test for Strength Properties of Adhesives in Shear by Tension Loading (Metal to Metal)
ASTM D1184	Method of Test for Strength of Adhesive Bonds on Flexural
ASTM D1304	Methods of Testing Adhesives Relative to Their Use as Electrical Insulation

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When a preproduction sample is required (see 6.2), it shall meet all the requirements of this specification.

3.2 Materials. The adhesive shall consist of a two-component system consisting of an epoxy resin and an activator.

3.3 Physical properties.(uncured).

3.3.1 Viscosity. The viscosity of the resin shall not exceed 11,000 centipoises (cps) for type I and 600 cps for type II at 23 plus or minus 2 degrees C (73.5 plus or minus 3.6 degrees F).

3.3.2 Appearance. The adhesive, in a mixed condition, shall be translucent and colorless.

3.4.3. Contamination. The resin and activator shall be free of dirt, foreign material or other contaminants.

3.3.4 Shelf life. The shelf life of the resin and activator in unopened containers shall be a minimum of 1 year from date of manufacture when stored at a temperature less than 27 degrees C (80.6 degrees F).

3.4 Physical properties (cured).

3.4.1 Specific gravity. The specific gravity shall be 1.2.

3.4.2 Tensile strength. The tensile strength shall be 8300 pounds per square inch.

3.4.3 Compressive strength. The compressive strength shall be 14,100 pounds per square inch.

3.4.4 Flexural strength. The flexural strength shall be 13,900 pounds per square inch.

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3.4.5 Tensile shear strength (aluminum to aluminum). The tensile shear strength shall be 1,600 pounds per square inch at 25 degrees Celsius.

3.4.6 Elongation. The elongation shall be 3.25 percent.

3.4.7 Operating temperature range. The operating temperature range shall be from minus 60 to plus 150 degrees Celsius (C).

3.4.8 Density. The density shall be 0.043 pounds per cubic inch.

3.4.9 Hardness. The hardness shall be Shore D 80 plus or minus 5.

3.5 Electrical properties.

3.5.1 Dielectric constant. The dielectric constant shall be 4.3 at 1,000 cycles.

3.5.2 Dielectric strength. The dielectric strength shall be 370 volts per Mil.

3.5.3 Volume resistivity. The volume resistivity shall be as follows:

At 25 degrees C. ohm-cm: 5.9×10^{13}

At 100 degrees C. ohm-cm: 5.0×10^{9}

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perf orm any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots. Preproduction samples which do not meet all the requirements of this specification shall be subject to rejection (see 3.1).

4.3 Classification of tests. The examination and testing of the compound shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample material and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examination and tests:

a. Inspection

b. Examination of product

c. Workmanship

4.4 Lot size and sampling.

4.4.1. Lot size. A lot size shall consist of all compounds manufactured at one time from one batch forming part of one contract or order, and submitted for acceptance at the same time and place.

4.4.2 Sampling. Unless otherwise specified (see 6.2) a sufficient quantity of compound shall be selected from each lot. Failure of the sample to meet the quality conformance requirements shall be cause for lot rejection.

4.5 Test methods.

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4.5.1 Uncured property tests.

4.5.1.1 Viscosity. The viscosity of type I shall be determined by using a Brookfield viscometer, model RVF, with a number 4 spindle at 4 revolutions per minute (rpm) or with equivalent equipment. The viscosity of type II shall be determined by using a Brookfield viscometer, model RVF, with a number 2 spindle at 20 rpm or with equivalent equipment. The viscosity shall be as specified in 3.4.1.

4.5.1.2 Appearance. The appearance of the mixed adhesive shall be determined visually. The adhesive shall be translucent and colorless.

4.5.1.3 Contamination. The resin and activator shall be visually examined to determine compliance with 3.4.3.

4.5.1.4 Shelf life. The resin and activator shall be tested for conformance with all the requirements specified herein after completion of the specified storage period. The supplier's certificate of compliance with 3.4.4 may be accepted during the storage period.

4.5.2 Cured property tests.

4.5.2.1 Test specimen preparation. The applicable test specimens shall be prepared from a compound that has been prepared in accordance with the manufacturer's recommendation (see 6.3).

4.5.2.2 Electrical property tests. Electrical property tests of the compound shall be as specified in Table I.

TABLE I

ELECTRICAL PROPERTIES		
CHARACTERISTICS	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Dielectric Constant	3.5.1	4.5.2.2.1
Dielectric Strength	3.5.2	4.5.2.2.2
Volume Resistivity	3.5.3	4.5.2.2.3

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4.5.2.2.1 Dielectric constant. The dielectric constant test shall be conducted in accordance with ASTM D 1304.

4.5.2.2.2 Dielectric strength. The dielectric strength test shall be conducted in accordance with ASTM D 1304.

4.5.2.2.3 Volume resitivity. The volume resistivity test shall be conducted in accordance with ASTM D257.

4.5.2.3 Mechanical property tests. Mechnical property tests of the compound shall be as specified in Table II.

TABLE II

MECHNICAL PROPERTIES		
CHARACTERISTICS	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Specific Gravity	3.4.1	4.5.2.3.1
Tensile Strength	3.4.2	4.5.2.3.2
Compressive Strength	3.4.4	4.5.2.3.3
Flexural Strength	3.4.4	4.5.2.3.4
Tensile Shear Strength	3.4.5	4.5.2.3.5
Visual Examination	3.4.2	4.5.1.2
Density	3.4.8	4.5.2.3.6
Hardness	3.4.9	4.5.2.3.7

4.5.2.3.1 Specific gravity. The specific gravity test of the compound shall be conducted in accordance with ASTM D901.

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4.5.2.3.2 Tensile strength. The tensile strength test of the compound shall be conducted in accordance with ASTM D638.

4.5.2.3.3 Compressive strength. The compressive strength test shall be conducted in accordance with ASTM D695.

4.5.2.3.4 Flexural strength. The flexural strength test shall be conducted in accordance with ASTM D1184.

4.5.2.3.5 Tensile shear strength. The tensile shear strength shall be conducted in accordance with ASTM D1002.

4.5.2.3.6 Density. The density shall be determined in accordance with FED-STD-406, Method 5011.

4.5.2.3.7 Hardness. Hardness shall be determined in accordance with FED-STD-406, Method 1083.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be levels A, B or C as specified (see 6.2).

5.1.1 Level A. Coating compound shall be furnished in cans conforming to type V, Class 2 of specification PPP-C-96. Exterior coatings, plan B and size-seam stripping shall be required. Like size cans shall be arranged and packed for overseas shipment in accordance with the appendix to specification PPP-C-96. Cans when packed in tiers, shall have fiber separator pads placed between the tiers. The fiber pads shall conform to grade 6 (W6C or W6S) of specification PPP-B-636.

5.1.2 Level B. Cans of like size shall be arranged and packed for domestic shipment in accordance with the appendix to specification PPP-C-96. When packed in tiers, cans shall have a 200 pound minimum bursting strength fiber pad placed between tiers. Fiber pads shall conform to Class 1 of specification PPP-B-636.

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5.1.3 Level C. Coating materials, quantity and packaging as specified (see 6.2), shall be packed in containers of the type, size and kind commonly used for the purpose and accepted by common carrier for safe delivery at destination. The shipping containers shall comply with the Uniform Freight Classification rules, or regulations of other carriers as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking rquired by the contract or order (see 6.2), interior containers, exterior shipping containers, and palletized unit 1 pads shall be marked in accorance with standard MIL-STD-129. Marking shall also include the following:

a. This specification number.

b. Type I or Type II.

c. Manufacturer's lot or batch number.

d. Manufacturer's number and material designation.

e. Purchase order number.

f. Date of manufacture.

g. Date of shipment.

h. Shelf life expiration date.

5.2.1 Warning marking. Each container shall be clearly and legibly marked with the following warning marking.

CAUTION

AVOID FUMES OR CONTACT WITH SKIN

6. NOTES

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6.1 Intended Use. The epoxy resin system covered by this specification is intended for bonding metal to metal, metal to plastic, plastic to plastic, and electronic components to boards or metal surfaces where excellent physical and electrical properties are required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Type I or Type II

c. Whether a pilot lot is required (see 3.1), and if so, pertinent details

d. Applicable levels of preservation, packaging and packing (see 5.1)

e. Quantity and size of containers

f. Precautionary handling marking required (see 5.2.1)

6.3 Mixing data. The compound shall be mixed in accordance with the manufacturer's recommendations.

6.3.1 Pot life. The pot life of compound when properly mixed shall (at 25 degrees centrigrade) be 30 minutes.

6.3.2 Cure time. The curing time of the compound when properly mixed shall be as follows:

At 25 degrees C.:	72 hours
At 65 degrees C.:	4 hours
At 100 degrees C.:	2 hours

6.4 Toxicology. Certain of the unreacted monomers and catalysts are known to be able to cause some toxic effects if no precautions are taken. Because of the range of possible effects among the chemicals involved, users should inquire to their suppliers concerning specific precautions applicable to the systems involved.

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6.5 Supersession data. This specification includes the requirements of Missile Interim Specifications MIS-14110A, dated 23 August 1965, MIS-13262, dated 5 September 1967, and MIS-12836, dated 7 February 1967.

Custodian:
Army-MI

Preparing Activity:
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